Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 12 · SLO 2 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 2 : Session â€“ 12 : SLO â€“ 2 | Implementation of Doubly Linked Lists â€” Concepts II

I. Fill in the Blanks

• To delete the last node, set the next pointer of the previous node to NULL.
• To delete a specific node, update the previous node's next pointer and the next node's PREV pointer.
• A doubly linked list can be traversed in both forward and BACKWARD directions.
• In C, the last node is identified when its next pointer is equal to NULL.
• In a doubly linked list, the head pointer points to the FIRST node.
• The previous pointer of the head node and the next pointer of the tail node should both be set to NULL.
• When freeing a node in C, we use the function FREE().
• A doubly linked list avoids the need for a temporary pointer for backward traversal because of the PREV pointer.
• While inserting after a given node, the new node's next pointer should point to the given node's NEXT (successor).
• In a doubly linked list, each node is connected in TWO directions.